Hyunwoo Kim – *Curriculum Vitae*

CONTACT

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RESEARCH INTERESTS

Quantum Machine Learning, Quantum Generative Models, Trainability and Expressivity of Variational Quantum Circuits, Pauli Propagation and Classical Surrogates, Quantum Error Mitigation, Industrial NISQ Algorithms

EDUCATION

Sogang University SEOUL, KOREA
Bachelor of Science (B.S.) in Physics & Artificial Intelligence Mar 2020 – Feb 2026

- GPA: 3.81/4.30 (95.8th percentile; cohort average 91.4); *Magna Cum Laude*
- External coursework at Yonsei University (credit exchange): Quantum Information Science, Quantum Computing
- Physics: Quantum Physics, Modern Optics, Statistical Physics, Electromagnetics, Solid-State Physics, Classical Dynamics, Computational Physics II
- AI & CS: Data Structures, Algorithms, Statistics, Linear Algebra, Machine Learning, Deep Learning

PUBLICATIONS & PRESENTATIONS

† Equal contribution.

Preprints

(4) Y. Lee†, **H. Kim**†, "Trainability and Mode Separation of Mixed IQP-QCBMs," (2026). arXiv:2607.27883
- Accepted, Technical Papers, Machine Learning to Advance Quantum Technologies (ML4QT) Symposium 2026, *Waterloo*, Sep 23–25, 2026

(3) **H. Kim**†, Y. Lee†, "SAFE ma-QAOA: Surrogate-Assisted and Fine-Tuning Enhanced Multi-Angle QAOA with Parameter Distillation," (2026). arXiv:2605.23377
- Accepted, Technical Papers Track, IEEE International Conference on Quantum Computing and Engineering (QCE) 2026, *Toronto*, Sep 13–18, 2026
- Invited talk, Waterloo Quantum Club (WQC), University of Waterloo, *Online*, Jul 15, 2026
- Poster, Quantum Future Korea 2026 (2nd National Undergraduate Quantum Conference), *Seoul*, May 15, 2026

Presented Work

(2) **H. Kim**†, Y. Lee†, "PADO-Pauli: A GPU-accelerated Implementation of Pauli Propagation Surrogate," (2026). github.com/Norma-Q/PADO-Pauli
- Poster, ML4QT Symposium, *Waterloo*, Sep 23–25, 2026
- Poster, IEEE QCE 2026, Poster Track, *Toronto*, Sep 13–18, 2026
- Poster, Conference on Quantum Information (Quantum Korea 2026), *Seoul*, Jul 1–3, 2026

(1) **H. Kim**†, Y. Lee†, "Two Approaches of Potential Quantum Advantage with Quantum Generative Models in Drug Discovery," (2026).
- Poster, 1st National Undergraduate Quantum Conference, *Seoul*, Feb 28, 2026
- Poster, Annual Meeting of the Quantum Information Society of Korea, *Seoul*, Feb 25–27, 2026

RESEARCH EXPERIENCE

NORMA, Inc., Quantum AI Team SEOUL, KOREA
Researcher Jul 2025 – Present

- Implemented quantum generative models (QGAN, IQP-QCBM) in Qiskit, PennyLane and NVIDIA CUDA-Q to study the expressive power and trainability of variational circuits
- Built PADO-Pauli, a GPU-accelerated (CUDA/ROCm), differentiable Pauli-propagation SDK; state of the art against pauli-prop (Qiskit), PauliPropagation (Julia) and cuPauliProp (NVIDIA), now extending to NPU backends
- Analysed noise resilience of VQE and QAOA on simulators and real hardware, and built a neural-network quantum error mitigation model for small-scale circuits

Pohang University of Science and Technology, Department of Physics — POHANG, KOREA
Research Assistant (Advisor: Prof. Yoonho Kim) — Jul 2024

- Aligned a high-precision interferometer and ran Bell-test measurements with a 1550 nm Kalmár laser
- Performed SPDC experiments for photon-pair generation and time-energy entanglement studies

Sogang University, Department of Physics — SEOUL, KOREA
Research Assistant (Advisor: Prof. Minsoo Kim) — Sep 2024 – Dec 2024

- Analysed quantum Hall transport in mesoscopic 2D systems, including valley-dependent transport and Dirac-cone phenomena

SELECTED PROJECTS

Quantum Advantage in Drug Library Generation by Quantum Machine Learning — SEOUL, KOREA
National Research Foundation of Korea, Quantum Computing Challenge — 2024 – 2026

- Built the evaluation pipeline for quantum-generated molecular drug libraries with KRAS G12D-binding ligand candidates as the benchmark target

Full-stack Remote Control System for Laboratory Instrumentation — SEOUL, KOREA

- Python system for remote digital-oscilloscope control: device control, data extraction, real-time visualisation and analysis

TEACHING & OUTREACH

NORMA, Inc., Conference Outreach — TOKYO & SEOUL
Technical Presenter — 2026

- Q2B 2026, Tokyo: ran the technical demo booth and presented the PADO-Pauli use case to industry researchers
- Quantum Korea 2026, Seoul: booth operation and technical presentation of quantum computing solutions

1st Quantum AI Competition (NORMA, Inc. & AI Factory) — JEONJU, KOREA
Technical Mentor — Aug 2025

- Mentored teams on quantum algorithm design and QML implementation (55 teams entered, 10 finalists)

Sogang University, Institute for Convergence Education — SEOUL, KOREA
Teaching Assistant, Introduction to AI Programming (Python) — Mar 2024 – Jun 2025

- Taught core Python and problem solving; prepared course materials and led debugging and practical sessions with the instructor

HONORS & AWARDS

Full Tuition Scholarship, Sogang University (merit-based, four years) — 2020 – 2025
Accommodation Scholarship, Lotte Scholarship Foundation (fully funded) — 2023 – 2025
Catholic Outstanding Student Scholarship, Sogang University — 2025
Sogang Alumni Association Scholarship, Sogang University — 2024

TECHNICAL SKILLS

Quantum SDKs: Qiskit, PennyLane, NVIDIA CUDA-Q, Quantinuum Nexus
GPU & HPC: CUDA, ROCm, GPU kernel optimisation, differentiable simulation backends
Programming: Python (NumPy, SciPy, PyTorch), C
Experimental: Interferometry, SPDC, Hall-effect transport

TRAINING

Seoul National University, The Basics of Quantum Computing and Information — Jul 2024 – Aug 2025
LG AI Institute, Advanced Machine Learning — Jan 2024 – Feb 2024

MILITARY SERVICE

Republic of Korea Army, Military Police — GANGWON-DO, KOREA
Sniper, Special Duty Team (SDT) — Feb 2021 – Aug 2022

- Counterterrorism and commander protection; managed and calibrated precision optical equipment and weaponry